

Parking Garage Counter System

Assignment 2: Software Project Team Formation and GitHub Collaboration

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Group 7

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Parking Garage Counter System

Section 1: Planning

Project Name

The name of this project is the Parking Garage Counter System.

Team Members

The members of Group 7 are José Leyva, Julio Mayedo, and William Riera. Each member contributed to the project's planning, design, and implementation, and a detailed description of each member's responsibilities is provided in the Team Responsibilities section of this document.

Project Description

The Parking Garage Counter System is a console-based Python application designed to help a parking garage monitor the usage of its parking spaces. The program manages a minimum of 25 parking space records. Each record stores a parking spot number, the license plate of the assigned vehicle, the vehicle type, and the current parking status. Through a text-based menu, an attendant can add new records, search for a specific space, display all spaces, update the status of a space, and generate a summary report that reflects the current state of the garage.

Problem Statement

Parking garages often rely on manual or informal methods to track which spaces are occupied, available, or reserved. These methods are time-consuming and prone to error, which can lead to overbooking, lost revenue, and frustrated customers. The Parking Garage Counter System addresses this problem by providing a simple, centralized tool that records the status of every parking space and produces an accurate summary on demand, allowing staff to make quick and informed decisions about space availability.

Project Goals

- Maintain accurate records for at least 25 parking spaces.
- Track each space using one of three status values: Occupied, Available, or Reserved.
- Provide core operations to add, search, display, and update parking space records.
- Generate a summary report containing four reporting items.
- Demonstrate collaborative software development using a shared GitHub repository.

Inputs

- Parking spot number, used as the identifier for each space.
- Vehicle license plate.
- Vehicle type (for example, car, motorcycle, or truck).
- Parking status (Occupied, Available, or Reserved).
- Menu selections that direct the program's actions.

Outputs

- A formatted list of all parking space records.
- Search results for an individual parking space.
- Confirmation messages after a record is added or updated.
- A summary report showing total parking spaces, occupied spaces, available spaces, and reserved spaces.

Section 2: Design

Program Structure

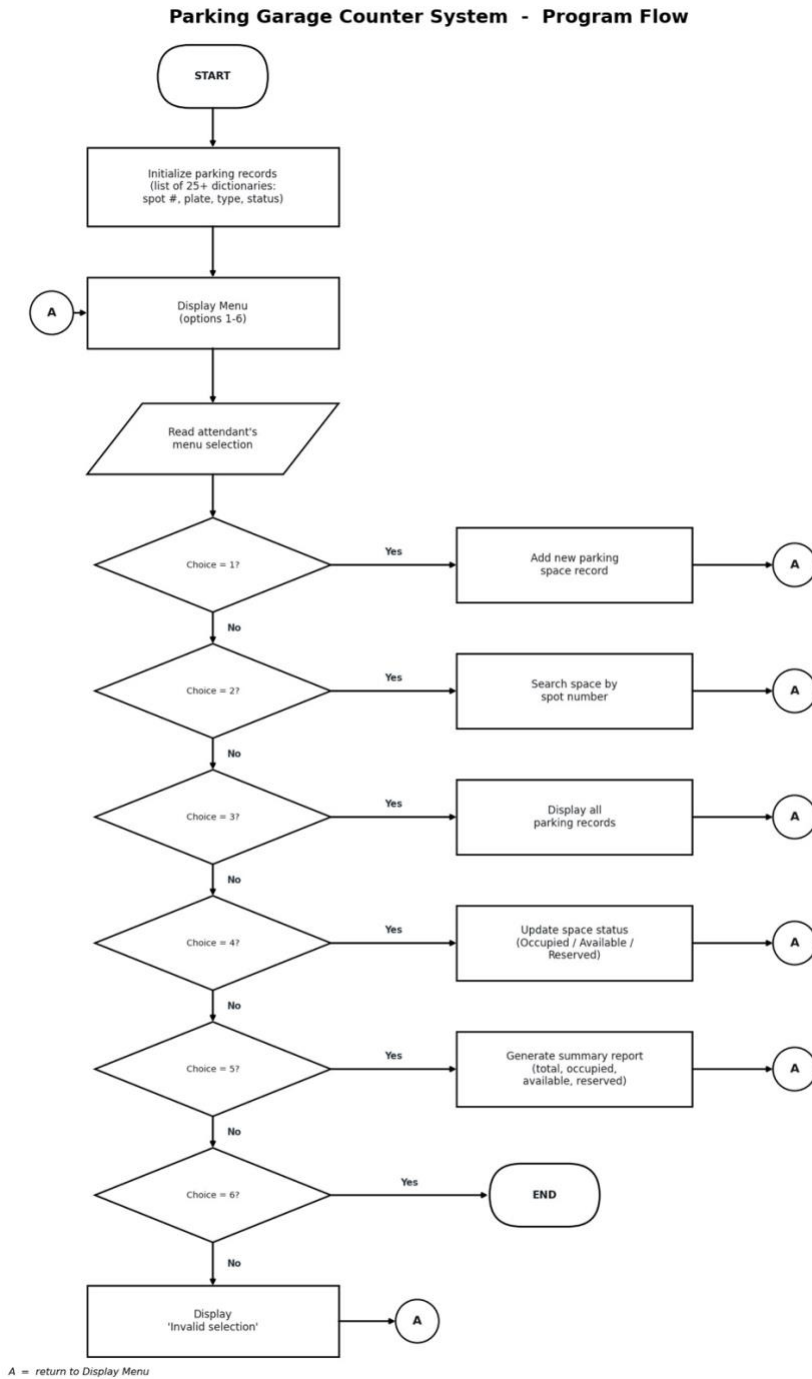
The application follows a menu-driven structure controlled by a main loop. When the program starts, it displays a menu of options and waits for the user to make a selection. Each selection calls a dedicated function that performs one of the program's six core operations. After an operation completes, the program returns to the menu until the user chooses to exit. Parking space records are stored in memory as a list of dictionaries, where each dictionary represents a single parking space and its associated data.

The program provides the following six functions, which are required of all projects:

1. Add Record – creates a new parking space record and adds it to the list.
2. Search Record – locates a parking space by its spot number and displays its details.
3. Display All Records – lists every parking space record currently stored.
4. Update Record Status – changes the status of a space to Occupied, Available, or Reserved.
5. Generate Summary Report – counts and displays the four reporting items.
6. Exit Program – ends the main loop and closes the application.

Flowchart

The flowchart below illustrates the overall flow of the program, from displaying the menu through each operation and back to the menu until the user exits.

Figure 1*Program Flowchart of the Parking Garage Counter System*

Team Responsibilities

The table below summarizes the roles and responsibilities of each team member.

Team Member	Role and Responsibilities
José Leyva	Role: Software Developer
	Planning: contributed to the project description, problem statement, goals, inputs, and outputs.
Julio Mayedo Fernández	Role: Software Developer
	Design: designed first draft of solution flowchart.
William Riera	Role: Project and Documentation Manager, GitHub Coordinator, Lead Software Developer
	Planning: prepared project description, problem statement, goals, inputs, and outputs.
	Design: designed final version of solution flowchart.
	GitHub Setup: repository creation, README, folder structure, and screenshots.
	Documentation: prepared solution document.

Section 3: GitHub Repository Setup

Repository

The team created a shared GitHub repository to host the project's source code and documentation. All team members were added as collaborators so that each person can commit their contributions directly. The repository URL is provided below and again at the end of this document:

<https://github.com/wrieraitpro/team7-parking-garage-counter-system>

README File

The repository includes a README.md file that introduces the project to anyone who views it. The README describes the purpose of the Parking Garage Counter System, lists the team members, summarizes the program's features, and explains how to run the application from the console.

Folder Structure

The repository is organized into a clear folder structure that separates source code, flowchart, documentation, and screenshots, as shown below.

```
team7-parking-garage-counter-system/  
├── README.md  
├── documentation/  
│   └── team7_parking_garage_counter.docx  
├── flowchart/  
│   └── parking_garage_counter_flowchart.jpg  
├── sourcecode/  
└── screenshots/
```

Section 4: Screenshots

The following screenshots document the GitHub repository and the team's collaboration.

Each screenshot should be inserted in place of the bracketed note beneath its figure label.

Figure 2

Repository Home Page

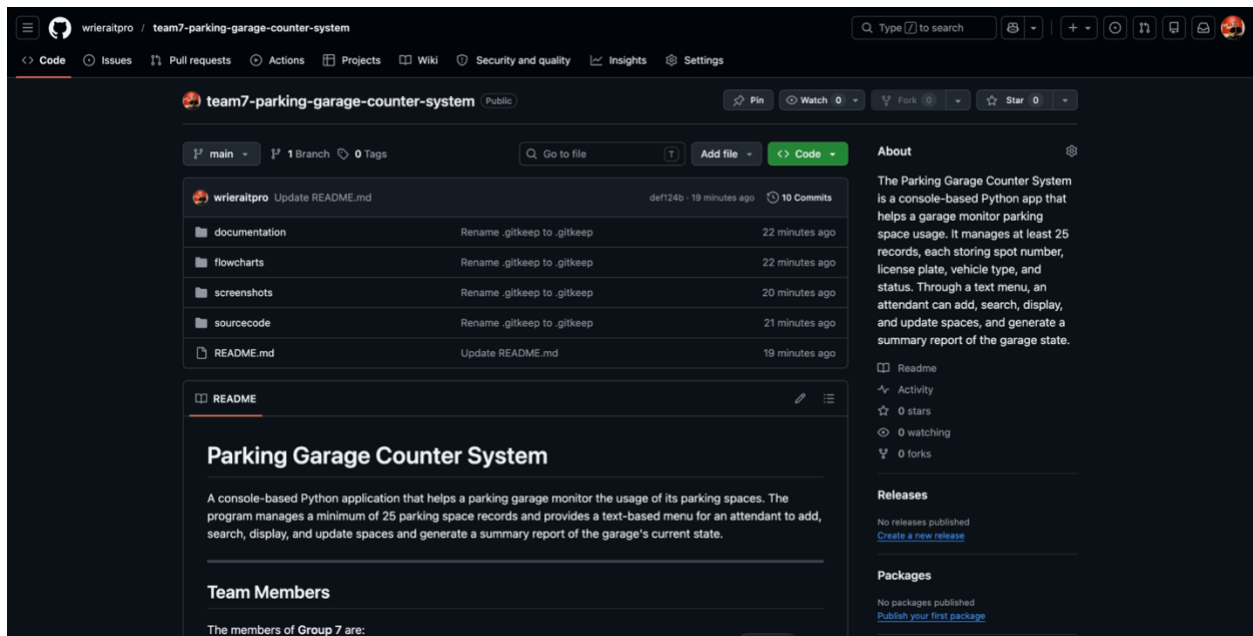


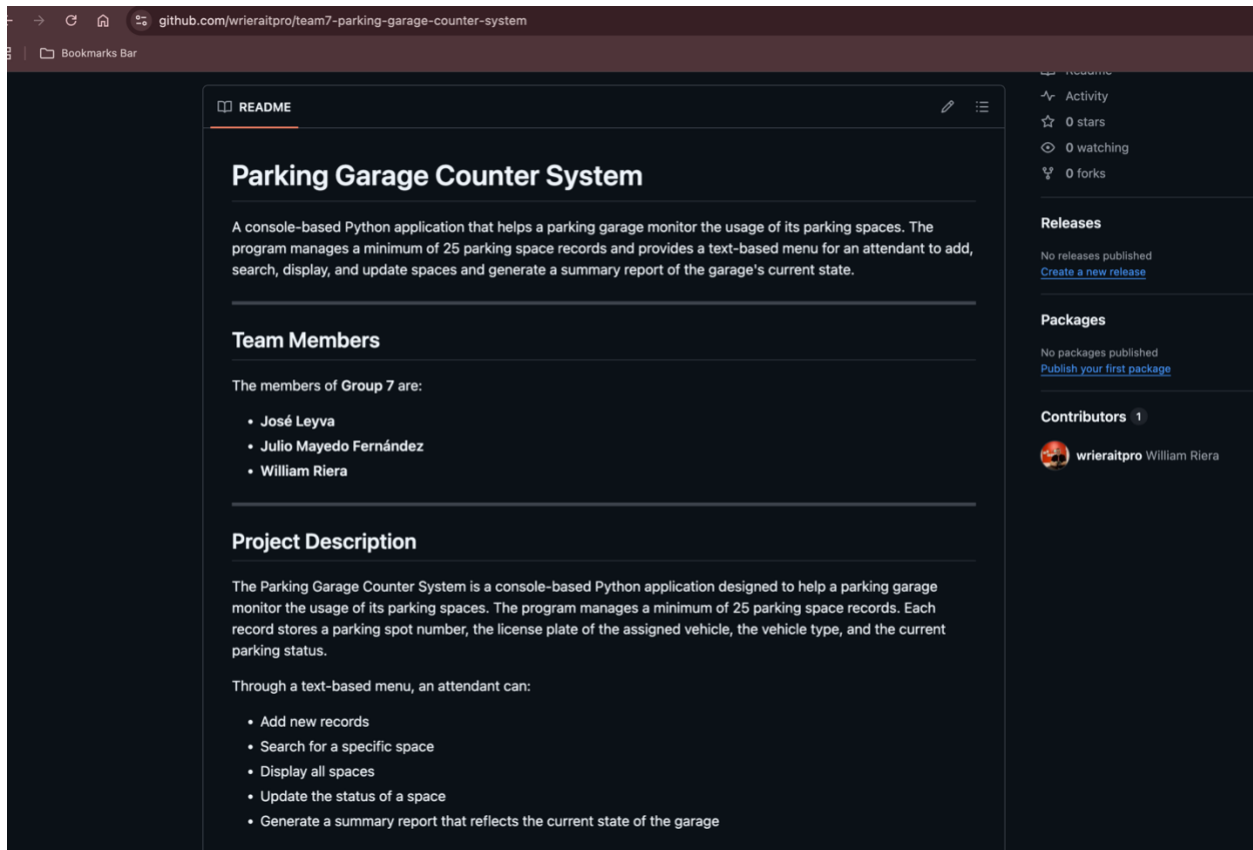
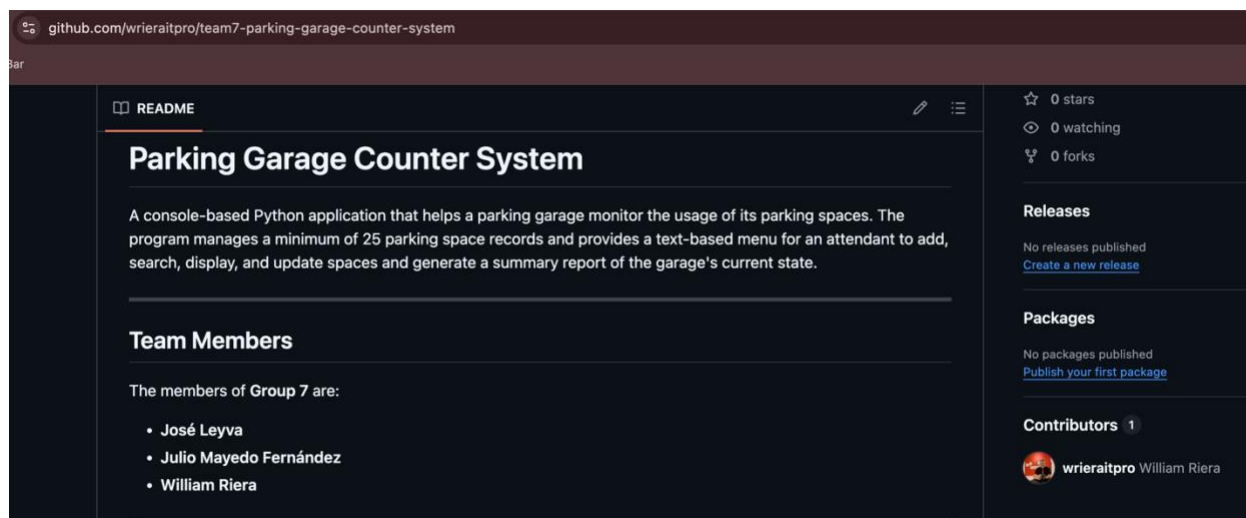
Figure 3*README File as Rendered*

Figure 4

Team Members (Collaborators as Planned)

**Figure 5**

Team Members – Collaborators Invitation

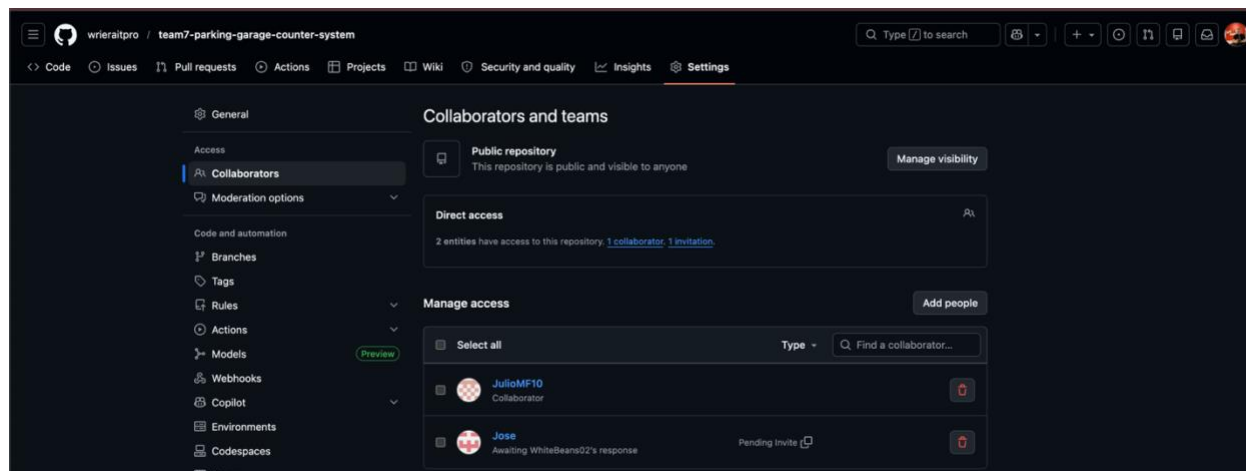


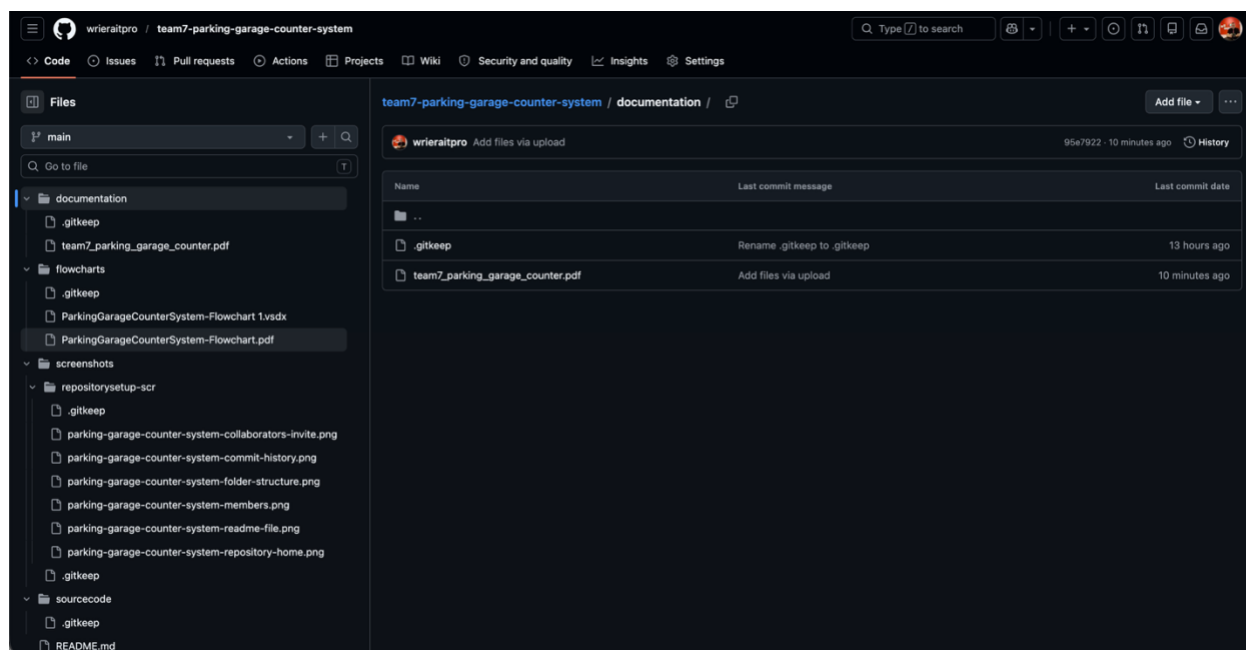
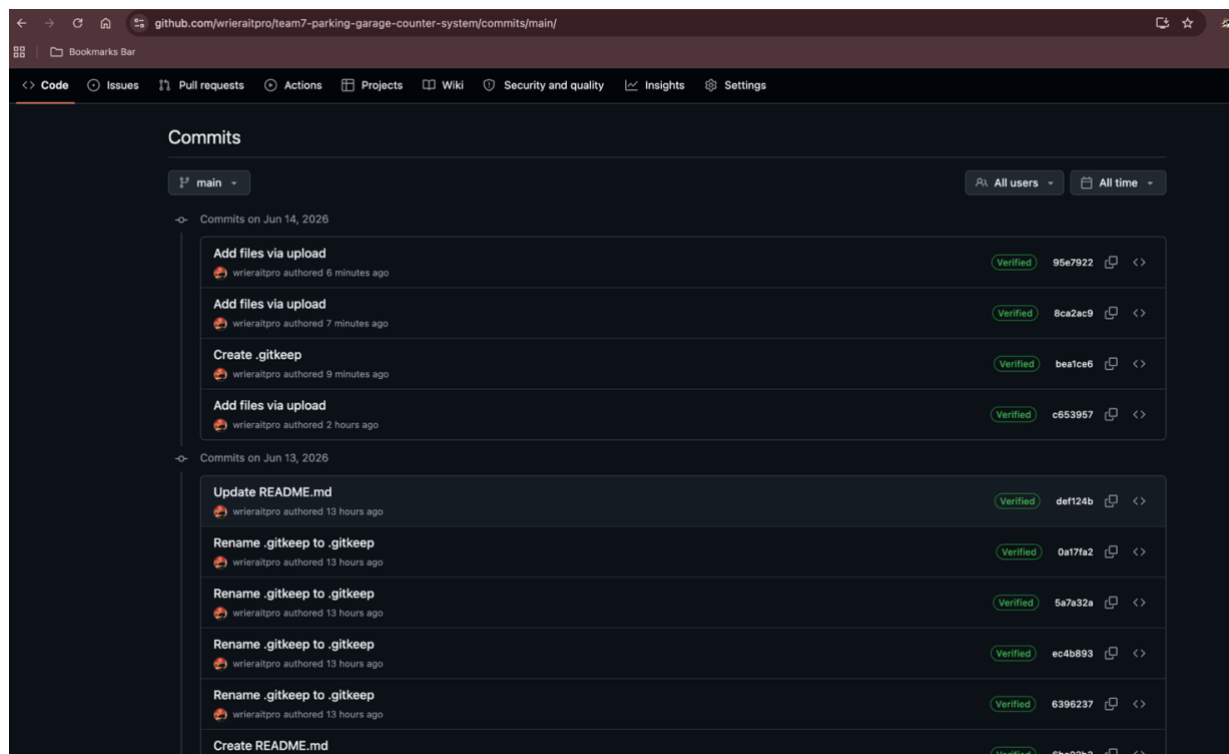
Figure 6*Folder Structure*

Figure 7*Commit History***GitHub Repository URL**

<https://github.com/wrieraipro/team7-parking-garage-counter-system/tree/main>

References

- GitHub. (n.d.-a). *About repositories*. GitHub Docs. Retrieved June 14, 2026, from <https://docs.github.com/en/repositories/creating-and-managing-repositories/about-repositories>
- GitHub. (n.d.-b). *Creating a new repository*. GitHub Docs. Retrieved June 14, 2026, from <https://docs.github.com/en/repositories/creating-and-managing-repositories/creating-a-new-repository>
- GitHub. (n.d.-c). GitHub Docs. Retrieved June 14, 2026, from <https://docs.github.com/en>
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